Yarmouk University

An Evidence Based Critical Review of a Three Level Active Learning Framework

CS603 | RESEARCH METHODOLOGIES

Supervised By:

Prof. Ahmad T. Al-Taani

Student:

Yaser Kh. Alomari
(St No. 2024751005)

Original article: Zhu, Y. and Zhang, J. (2026) "Addressing representation instability, strategy rigidity, and class imbalance in active learning for agricultural pest and disease recognition", Computers and Electronics in Agriculture, 250, 111938. https://doi.org/10.1016/j.compag.2026.111938

Central judgement

In this article, there is an effective and well-tested enhancement for an active learning baseline algorithm with dual branches. The best piece of evidence that supports it is the repeated run ablation test conducted on two agriculture recognition benchmarks. However, the claims regarding robustness to the class imbalance problem, reduced annotation cost, and application to the field of agriculture diseases go beyond what is directly supported by the experiments on these particular balanced benchmarks with low resolution and single labels (Zhu and Zhang, 2026, pp. 8–12)

At a glance evaluation

Research type / audience	**Experimental computer vision and active learning research; aimed at ML and smart agriculture researchers.**
Principal claim	Three coupled components improve selection: dynamic prototype memory, contextual bandit allocation and class balanced scores.
Main support	Two datasets; conventional AL baselines; five run means; 2^3 ablation combinations; parameter sensitivity; GAN comparison; computational overhead.
Main reservation	Both datasets are balanced and use 84 × 84 inputs; disease specific, multi label, naturally long tailed and real annotation cost outcomes are not tested.
Overall verdict	Relevant contribution with good controlled evidence for these datasets, but external validity and some design/reporting details need strengthening.

Evidence convention: "reported" means explicitly present in the supplied article; "not clearly reported" identifies a transparency gap rather than proving that an action was never performed. Numerical differences are in percentage points (pp), not relative percent.

1. Paper classification, purpose and intended audience

This research paper does not belong to a literature review or case study of field application. The main goal of this article is to solve the problem of the labeling bottleneck in agricultural image classification by overcoming three shortcomings of the current active learning loop: noisy class representatives, fixed exploration exploitation rate, and biased distribution of selected labels. The target audience of this research includes the scholars of computer vision, active learning, and digital agriculture (pp. 1–3). Technical terminology, formulas, Algorithm 1, and experiments are appropriate for this audience. The title is focused on pests and diseases but the testing task concerns insects and weeds, and there is no evidence of plant disease detection (pp. 7–8).

2. Problem significance and scientific claims

This line of reasoning seems very compelling, in that labels of agricultural products might require certain expertise, hence the necessity of intelligent allocation of a constrained annotation budget (pp. 1–2). The paper aligns three elements of the proposal with three clear limitations, which allows one easily to follow the research logic of this article. Its key assertions are: prototype EMA stabilization increases feature stability, contextual bandit increases selection ratios, and inverse frequency scoring helps mitigate acquisition-induced class skew (pp. 2–3).

Crucial difference: This paper evaluates prediction accuracy in relation to the availability of labeled data, but not annotator hours, expert labor costs, or implementation in the field. Hence, it is quite possible that labeling need would be decreased, but the decrease in costs as an operational result cannot be empirically validated (Abstract; pp. 9, 12). Recommendation: report annotation time, expert disagreement and compute cost adjustment along with budget curves.

3. Literature context and originality

Related Work consists of Uncertainty Sampling, CoreSet, BADGE, Learning Loss, semi-supervised FixMatch, prototype learning and agricultural augmentation (pp. 2-4). This range is justified. The novel contribution consists primarily of the composition and modification of existing parts – EMA prototypes, contextual bandit algorithm on selection ratio, inverse frequency weighting during acquisition (p. 3). The authors correctly indicate a two-branch algorithm as the baseline for comparison (pp. 3-4).

Originality would be more convincing if the direct comparison with other adaptive or class-balanced algorithms was stated on the formulation level. The difference between the “newly developed framework” and “never before used underlying mechanism” is important since EMA, bandits, and inverse frequency weighting existed prior to the article. The review of literature is relevant, but alone does not prove uniqueness or superiority of the approach in agricultural Active Learning.

4. Methodological validity: how sample selection works

4.1 Dynamic Prototype Memory (DPM)

The Equation 6 ensures that the prototype of every class remains an exponential moving average of its previous prototype and its estimate in the current round. A proof of variance reduction and noise reduction is provided by Proposition 1 on pages 4-5. It seems to be a reasonable stabilization procedure, and the accuracy of the IP06 model improves from 68.22% to 70.85% (+2.63 pp; Table 2 on page 8) due to DPM only ablation when there is

90% availability. Nonetheless, the backbone model gets retrained between rounds, which results in potential variation of embedding spaces. The theoretical bound relies on the stability of the underlying structure.

4.2 Context Aware Sample Selection (CtxB)

The neural contextual bandit selects K from nine possible values ranging from 0.1 to 0.9 depending on the training progress, validation accuracy, accuracy change, and labeled class distribution as the context (Equation 7; Fig. 3, p. 6). This is an immediate solution to static branch weighting. For IP06, CtxB, when combined with the 90% availability, achieves a 74.37% classification performance improvement from 68.22% without learning (+6.15 pp; Table 2, p. 8). Yet, the learned policy gets one delayed reward at each step, as noted in Section 6 (p. 12). Given few actions, there can be a lack of stability of the policy; it would be instructive to see comparisons with the policy of selecting fixed/scheduled K values.

4.3 Class Balanced Scoring (CB)

The above equations (8)(9) assign weights to candidates using inverse proportions of labeled class frequencies and predictions for candidate classes. It is appropriate theoretically since there are no true labels until the query stage, and the authors clearly differentiate between induced class imbalance due to selection and inherent class imbalance of the data set (pp. 6, 8). The approach does not involve any ground truth candidate labels in its scoring criteria. But the argmax predicted classes can be wrong at an earlier stage in learning, and alpha=0.5 moderates but does not prevent errors in prioritizing candidates. Class based precision and recall will assess the CB effect on the target class.

4.4 Selection batch accounting

Equation 4 and Algorithm 1 specify the selected batch as the union of two branch selected sets (p. 4 and p. 7). In case a candidate is selected from both branches, the simple set union may lead to having less than the required b number of distinct images. The pseudocode printed does not provide details about preventing duplicates and adding to meet the exact value of b. It is an issue in implementation, and not an example of violating the budget.

5. Experimental design and execution

The authors utilize IP06: 6,000 images of insect pests (six classes, 1,000 each; 90/10 train/test) and Natural CW-10: 25,000 images of weeds (ten classes, 2,500 each; 80/20 train/test). The two datasets are balanced by definition. The inputs are scaled to 84 × 84 pixels. The primary network is a randomly initialized ResNet-18 that is trained with SGD for 100 epochs in a round; Active Learning protocol starts with 10% labels and grows by 10% in each round. The FixMatch also relies on unlabelled examples (pp. 7-8). This information is important for understanding the comparisons being made.

A key limitation of the approach derives from the design: balanced original datasets can exhibit the selection-induced skew but they cannot prove the robustness of a model to naturally very skewed distribution. Furthermore, six pest classes were chosen in IP102 since each class has at least 1,000 images (p. 7). A long tail benchmark with very rare species is required in order to confirm the more ambitious claim of imbalance (p. 12).

Experiment description provides train/test split, while the description of the contextual bandit mentions the validation accuracy as part of the context and reward (pp. 6, 8). The article does not specify how the separate validation partition is generated and isolated from the final test report. This issue needs to be clarified before one can audit the independence of the test set; it is not evidence of the data leak. Ten rounds and the curves up to 90% of labels need to be explained in terms of budget ledger with the explanation of the last round and possible 100% of label availability step (pp. 8-10).

6. Results: what was demonstrated?

Under 90% data availability, the complete framework attains 83.36% on IP06 and 85.07% on Natural CW-10. The respective dual branch ablation baselines obtain 68.22% and 71.62%, giving rise to improvements of +15.14 pp and +13.45 pp (Tables 2–3, pp. 8, 10). It is important not to confuse these differences between the baseline and the full framework with the smaller improvements over the best alternative published method at 20% availability: +4.12 pp on IP06 and +2.68 pp on CW-10 (Abstract; Figs. 7–8, pp. 9–10).

In particular, the 2×2×2 ablation layout of the complete design demonstrates how individual modules and all possible pairs lead to improvements before the full combination becomes better (Tables 2–3). This makes it evident why combining the mechanisms might be valuable under the conditions considered in this work. Nevertheless, gains from factorials do not guarantee the independence of effects for each of the mechanisms, especially in cases when selection histories and model states are different for multiple trials.

The GAN experiment is informative: Ours + GAN gives 84.91% on IP06 and 86.42% on CW-10, while Ours alone achieves 83.36% and 85.07% (Table 4, p. 12). The quality gate selects only generated images assigned by a pretrained classifier to the target class with confidence ≥ 0.8 (p. 12). While this is classifier consistency check, and not a biological one, independent verification by an agronomist is required to make claims about the realness of the symptoms/pests.

Figure captions and comparative curves make it easy to interpret the results; nevertheless, in most cases, the authors use overall accuracy, while the mechanism used is related to class imbalance. Per class recall, macro F1, balanced accuracy and class distribution in labeled pools would match the evidence with the mechanism much closer.

7. Statistical evidence, reliability and reproducibility

Tables 2-4 indicate mean accuracy from five independent runs, with ± values in ablation tables (pp. 8, 10, 12). This is more significant than a single trial and should receive

explicit recognition. However, table legends lack clear indication if ± represents standard deviation, standard error, or another dispersion measure. The supplied article lacks paired statistical significance testing, confidence intervals or effect size statistics for rankings presented. Therefore, numbers confirm the consistency of repeated runs, but statistical significance is not clearly stated.

The article provides model type, optimizer, pseudo label threshold, hardware configuration and important hyperparameter values (p. 8). Data Availability statement says “Data will be made available on request” (p. 13). The paper does not provide code availability, publicly available split and manifest files or links to them in the supplied text. Code, seeds, splits, chosen IDs per round and model checkpoints availability could improve independent reproduction. “Not reported” is the correct choice here, as we cannot assume the absence of code.

One particular inconsistency in writing can be found in Section 4.3 (p. 10), where +6.15 pp at high data availability is called “larger” than +6.92 pp at low data availability. Numerically, $6.15 < 6.92$. It does not deny the contribution of CtxB model to improvements, but requires correction and reconsideration of the trend.

8. Strengths, limitations and possible bias

Demonstrated strengths

- The problem statement is very clear and the proposed three mechanisms directly address some of the limitations (pp. 1-3).
- The evaluation process uses two agriculturally relevant benchmarks, well-constructed baseline and full combination module ablation study (pp. 8-10).
- Sensitivity analysis and GAN complementarities have been validated; not mentioned as future work (pp. 10-12).
- The results are based on five different runs and computation time overheads have also been calculated: 8.5 to 8.9 minutes per run (+4.7%) (Table 5, p. 12).
- The balanced data, bandit-based feedback and low-resolution limitations are acknowledged (Section 6, p. 12).

Material qualifications

- External validity: the effect of naturally unbalanced agricultural classes and the effect of higher resolution crop diseases were not considered (pp. 7–8, 12).
- Outcome coverage: overall accuracy cannot directly confirm minority class sensitiveness or any benefit to manual labeling (pp. 9–12).
- Clarity of validation: the validation policy is context-based but there is no explicit definition of the validation partition (pp. 6, 8).
- Clarity of implementation: the overlap of branch decisions and the final budgets are not entirely defined in the published procedure (pp. 7–8).
- Reporting: uncertainties, dataset IDs and code/splits manifest could be more explicitly described (pp. 8, 10, 13).

Assessment of Bias: the research paper has stated its source of funding and declared no conflict of interest (pp. 12–13). There seems to be no sign of author's misbehavior or bias on their part. A better case for consideration is selection bias while evaluating where selection bias may have favored results indicating rebalancing of the portfolio in light of the six well populated categories.

9. Presentation, terminology and references

Structure – The article has an organized structure of Introduction, Related Work, Methodology, Experiments, Conclusions, Limitations and References (pp. 1-13). Figures 1 to 4 show a logical illustration of the process. Tables 2 to 5 offer quantitative data. Terms DPM, CB, CtxB, EMA and GAN are defined along with their explanations. Primary literature references for ResNet, FixMatch, CoreSet, BADGE, prototypical learning etc., have been cited in the article (pp. 2-4, 13). This is positive as per the evaluation criteria set by the instructor.

Table 5 comparison mentioned above is in need of copyediting. The 8.5 vs 8.9 min/round comparison shows incremental runtime overhead, not a total cost of annotation or deployment. An improved article would have kept observations on computational timings apart from the economics of human labeling (p. 12).

10. Recommendations and referee style verdict

1. Generalizability: Investigate generalization to the original long tailed IP102, disease-specific datasets, higher-resolution field imagery, and multi-label disease co-occurrence.
2. Evidence of direct class balancing: Document the selected label histograms per round, the minority class recall, the macro-F1, balanced accuracy and calibration per class.
3. Rigorous evaluation protocol: Document the explicit train/validation/test manifests, protect the bandit rewards from the ultimate test performance, and resolve the round/budget accounting discrepancy.
4. Reproducible acquisition process: Document the overlap avoidance between branches, individual budget refilling, random seeds, queried image IDs and the bandit action sequence.
5. Robust inferential procedure: Offer repeated evaluations per random seed pair, confidence intervals, uncertainty metrics and suitable statistical tests on the main differences.
6. Label acquisition efficiency: Quantify the annotation time by experts, validation of generated images, training/scoring expenses and total cost relative to a specific accuracy target.

Conclusion. This study is methodologically valuable and convincing and contains empirically meaningful support in light of its two balanced agricultural classification metrics. The ablations are its most convincing proof. External validity, the difference between label efficiency and labeling cost, missing class sensitivity information, and a handful of protocol/reporting confusions represent the main scientific concerns regarding it. Being a published article, it should be reviewed retrospectively, not as a paper to be accepted; its primary findings can be used, but not its broader applicability claims.

Evidence audit: critical observations mapped to the article

Critique point	Evidence location	Assessment
Claim: three level method	Abstract pp. 1–2; Fig. 1 p. 5	Confirmed: DPM + CtxB + CB
Theoretical prototype benefit	Eq. 6 and Proposition 1 p. 4; commentary p. 5	Conditional proof; embedding drift deserves testing
Predicted label class weighting	Eqs. 8–9 p. 6	Confirmed; risk from erroneous predictions is a critique
Batch union details	Eq. 4 p. 4; Algorithm 1 p. 7	Ambiguity: overlap/backfilling not specified
Balanced benchmarks	Section 4.1 pp. 7–8	Confirmed: 1,000 per class; 2,500 per class
Mean accuracy / runs	Tables 2–3 pp. 8, 10	Confirmed: 5-run means and ± values
Accuracy improvement	Tables 2–3 pp. 8, 10	Confirmed: 83.36 and 85.07 at 90%
GAN combination	Table 4 and Section 4.5 p. 12	Confirmed: 84.91 and 86.42
Runtime overhead	Table 5 p. 12	Confirmed: 8.5 to 8.9 min; +4.7%
Declared limitations	Section 6 p. 12	Confirmed: balanced data, sparse feedback, 84×84 images
Data availability	p. 13	Confirmed: “available on request”

The assessment follows the instructor’s questions about paper type, purpose, claims, evidence quality, bias, structure, clarity, references, terminology, validity of approach and execution (Al-Taani, 2026, Critical Analyses of a Literature, slides 29, 33–34, 39–45).

References (Harvard style)

Al-Taani, A.T. (2026) Critical Analyses of a Literature. CS603 Research Methodologies lecture slides. Yarmouk University. Unpublished teaching material.

Al-Taani, A.T. (2026) Presentation Skills: How to Create and Deliver an Effective Presentation. CS603 Research Methodologies lecture slides. Yarmouk University. Unpublished teaching material.

Zhu, Y. and Zhang, J. (2026) 'Addressing representation instability, strategy rigidity, and class imbalance in active learning for agricultural pest and disease recognition', Computers and Electronics in Agriculture, 250, 111938. doi:10.1016/j.compag.2026.111938.